IES Files, Fixed Parameters, Beam Angle, and the Full Calculation Pipeline

Overview

LuxScaleAI uses two sources of photometric data for each fixture:

1. **IES files** — real measured light distribution data (manufacturer or lab-measured)
2. **Fixed catalog parameters** — simplified constants (efficacy, nominal beam angle) used when IES data is unavailable or as a starting estimate

Understanding the difference between these two, and how beam angle flows through the entire calculation, is essential to interpreting any result from the tool.

Part 1 — Fixed Catalog Parameters

What they are

Fixed parameters are the simplified constants stored in `luxscale/lighting_calc/constants.py` and the fixture catalog JSON files. They describe a fixture in broad strokes:

Parameter	Example value	Where it lives
Efficacy (lm/W)	Interior: 110, Exterior: 145 / 160 / 200	`constants.py → led_efficacy`
Nominal beam angle	120°	`constants.py → beam_angle`
Power options (W)	`[9, 10]` for SC downlight	`constants.py → interior_luminaires`
Shape / size	Circle 0.464 m diameter	`constants.py → luminaire_shapes`

How fixed parameters are used in calculation

Step 1 — Rated flux per fixture:

```
Φ_rated = Power (W) × Efficacy (lm/W)
```

Example: 100 W × 145 lm/W = 14,500 lm

Step 2 — Effective flux after maintenance:

```
Φ_eff = Φ_rated × Maintenance Factor (MF)
```

Default MF = 0.63 (accounts for dirt, lamp depreciation, aging)

Step 3 — Lumen method average illuminance:

```
E_avg = (N × Φ_rated × MF) / Area
```

Where N = number of fixtures, Area = floor area in m^2

Step 4 — Minimum fixture count:

```
N_min = floor((E_required × Area / MF) / Φ_rated) + 1
```

This gives the starting point for the search loop.

Limitations of fixed parameters alone

When only fixed parameters are used (no IES file available):

- The calculation only produces **average lux** — no uniformity (U_0 or U_1)
- Beam angle is the nominal catalog value (120° default), not the real distribution
- No hot spots, wall illuminance, or work-plane grid are computed
- The result row will show `"Lux compliance basis": "lumen method (no IES grid)"`

Part 2 — IES Files (LM-63 Format)

What an IES file contains

An IES file (Illuminating Engineering Society format, LM-63 standard) stores the **measured luminous intensity distribution** of a real fixture in 3D space. The key data inside:

```
TILT=NONE
1  14500  1.0  37  1  1  1  -0.464  -0.464  0
...
0  5  10  15  20  25  30  ...  (vertical angles)
0  45  90  135  180  225  270  315  (horizontal angles)
14200  14100  13800  13200  12400  11500  10400  ...  (candela values)
```

Field	Meaning
Lumens per lamp	Total luminous flux from the fixture header
Multiplier	Scale factor (usually 1.0)
Vertical angles	Angles from nadir (0°) to horizontal (90°) or zenith (180°)
Horizontal angles	Azimuth planes around the fixture (0°–360°)
Candela values	Luminous intensity in each direction (candela)

How LuxScaleAI reads IES files

The tool uses `ies-render/module/ies_parser.py` (loaded dynamically via `importlib`) to parse LM-63 files. For performance, parsed data is cached via `lru_cache` — the same IES file is only parsed once per session.

A fast path also exists via `photometry_ies_adapter.py` which loads pre-processed JSON blobs from the catalog index, avoiding repeated file parsing.

What IES files enable that fixed parameters cannot

Capability	Fixed params only	With IES file
Average lux	✓ (lumen method)	✓ (grid-calibrated)
Uniformity U_0	✗	✓ (point-by-point grid)
Uniformity U_1	✗	✓
Real beam angle	✗ (nominal 120°)	✓ (half-power calculation)
Min/max grid lux	✗	✓
Asymmetric distribution	✗	✓
Wall and ceiling illuminance	✗	✓ (future)
Fixture-specific comparison	✗	✓ (each product different)

Part 3 — Beam Angle: Definition, Calculation, and Effect on Results

Definition

The **beam angle** (also called full beam angle) is the cone angle within which the luminaire emits at least 50% of its peak intensity. It is the angle between the two points on the intensity distribution where intensity drops to half the peak value (half-power points), measured as the full included angle.

```
Full beam angle = 2 × (angle from nadir to half-peak point)
```

How LuxScaleAI calculates beam angle from IES data

The calculation is done in `ies_fixture_params.py → approx_beam_angle_deg_for_horizontal()`:

1. Read the candela values for one horizontal plane slice
2. Find the peak candela value
3. Find where the candela drops to 50% of peak (half-power threshold = 0.5)
4. Linear interpolation between adjacent angle samples to find the crossing point
5. Multiply the half-angle by 2 to get the full beam angle

For asymmetric fixtures (e.g., asymmetric flood lights), this is done across all horizontal planes and the **narrowest** beam angle is reported as the primary value. This is conservative — it represents the tightest dimension of the beam.

```
# Simplified from ies_fixture_params.py
peak = max(candela_values)
cutoff = 0.5 * peak
# Find crossing point between consecutive angles
# Return 2 × half_angle as full beam angle
```

IES angle sign convention

IES LM-63 vertical angles follow a signed convention where angles can extend past 90° (toward the ceiling for up-light components). Some fixtures produce negative signed angles in the half-power calculation. LuxScaleAI always takes the absolute value for display:

```
def _non_negative_beam_angle_deg(v):
    return abs(float(v))
```

The result row includes a note when this correction was applied: `"Beam angle note": "IES vertical angles follow a signed convention..."`

How beam angle affects the calculation

1. Fixture filtering during the search

In `_uniformity_fallback_sweep_rows()`, narrow-beam fixtures are filtered out when high uniformity is required:

```
if beam_deg is not None and required_uniformity >= 0.62 and beam_deg < 34.0:
    continue  # Skip narrow-beam fixtures for high-uniformity spaces
```

Narrow beams create concentrated hot spots rather than even coverage. A 20° spot cannot produce $U_0 \geq 0.62$ in a normal room without extreme fixture density.

2. Work-plane grid illuminance (inverse-square law)

For each fixture, at each grid point, the illuminance contribution is:

```
E = I(θ_h, θ_v) × cos(i) / R²
```

Where:

- `I(θ_h, θ_v)` = candela from IES at the angle toward the sample point
- `cos(i) = (ceiling_height - work_plane_height) / R` — oblique incidence factor
- `R` = 3D distance from fixture to sample point (meters)

The candela value `I` comes directly from the IES angle table. A fixture with a narrow 30° beam has very high candela in the central cone but near-zero candela at wide angles. This produces:

- Very high illuminance directly below each fixture
- Very low illuminance between fixtures
- Low U_0 (high ratio of E_max to E_min)

A fixture with a wide 120° beam distributes light more evenly, increasing overlap between fixtures and raising the minimum illuminance relative to the average.

3. Flux scaling between IES and design lumens

```
I_used = I_file × (Φ_design_per_slot / Φ_IES)
```

If the IES header lumens differ from the design lumens (power × efficacy), candela is scaled proportionally. The ratio is capped between 0.25 and 4.0 to prevent meaningless results from bad IES headers:

```
ies_lumen_to_design_ratio_min = 0.25
ies_lumen_to_design_ratio_max = 4.0
```

4. Compliance decision

After the IES grid runs, compliance uses the grid average (E_avg_grid_lx) rather than the lumen-method average when available:

```
def _avg_lux_for_compliance(row):
    eg = row.get("E_avg_grid_lx")
    if eg is not None and float(eg) >= 0:
        return float(eg)
    return float(row.get("Average Lux") or 0.0)
```

The grid average is more accurate because it accounts for the actual photometric distribution, not just total lumens divided by area.

Part 4 — Full Calculation Pipeline Step by Step

Step 0 — Input parsing

```
sides=[L1, W1, L2, W2], height, place or standard_ref_no
```

- Room area computed via Brahmagupta formula (supports non-rectangular quadrilaterals)
- Length = max(side1, side3), Width = max(side2, side4)
- Zone determined: interior (height < threshold, default 5m) or exterior

Step 1 — Luminaire options

Based on zone, `determine_luminaire(height)` returns available fixture families from the merged IES catalog. Interior: downlight, triproof, backlight. Exterior: highbay, flood, street.

For compact rooms (area ≤ 400 m^2 or min dimension ≤ 14 m), triproof is prioritized first.

Step 2 — For each (luminaire, power, efficacy) combination

2a — Lumen method minimum:

```
Φ = power × efficacy
N_min = ceil((E_required × area / MF) / Φ)
```

2b — IES file resolution: `resolve_ies_path(luminaire_name, power_w)` looks up the IES file from the fixture map JSON. If found, `ies_params_for_file()` extracts beam angle, lumens from header, and stores the path for grid use.

2c — Main search loop (N = N_min to N_max):

For each fixture count N:

1. Check minimum spacing: `min(L/nx, W/ny) ≥ 0.8 m`
2. Compute lumen-method average: `E_avg_lm = N × Φ × MF / area`
3. Stop if `E_avg_lm > 1.35 × E_required` (over-lighting cap)
4. Skip if `E_avg_lm < E_required`
5. For each valid grid layout `(nx, ny)` where `nx × ny = N`:
   - Run IES point-by-point grid (`compute_uniformity_metrics`)
   - Compute U_0 = E_min / E_avg_grid
   - Compute compliance (lux gap, U_0 gap)
   - Keep best layout (minimum U_0 gap, then lux gap)

2d — Uniformity grid calculation (`compute_uniformity_metrics`):

Places a G×G sample grid on the work plane (default G determined by room size). For each of the `nx × ny` fixture positions, for each of the G^2 sample points:

```
1. Compute 3D distance R from fixture to sample point
2. Compute vertical angle θ_v = atan2(horizontal_dist, delta_z)
3. Compute horizontal angle θ_h = atan2(dy, dx)
4. Look up I(θ_h, θ_v) from IES candela table (nearest H, interpolated V)
5. Scale I by (design_lumens / IES_lumens)
6. E_contribution = I × cos(i) / R²
7. Add to total E at this sample point
```

Sum over all fixtures, then compute:

```
U₀ = E_min / E_avg_grid
U₁ = E_min / E_max
```

Step 3 — Compliance sorting

Results are sorted by:

1. Fully compliant (lux ✓ AND U_0 ✓) before non-compliant
2. Among compliant: `least_fixture_count_compliant` selected as primary
3. Remaining compliant options sorted by power efficiency
4. Non-compliant closest options included if no compliant found

Step 4 — Fallback sweep

If no standard layout meets both lux and U_0, `_uniformity_fallback_sweep_rows()` sweeps upward in fixture count with a relaxed lux cap (×1.65 instead of ×1.35), evaluating up to 300 IES grid calls to find layouts that improve U_0.

Step 5 — Output assembly

Each result row contains:

- Lumen method: Average Lux, Fixtures, Spacing X/Y, Total Power, Efficacy
- IES grid: E_avg_grid_lx, E_min_grid_lx, E_max_grid_lx, U_0_calculated, U_1_calculated
- Beam data: beam_angle_deg (from IES), beam_angle_nominal_deg (120°), Beam source
- Compliance: Standard margin (lux %), Standard margin (U0 %), Lux gap, U0 gap
- Meta: IES file, IES lumens, Layout grid, Selection method, Maintenance factor

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Part 5 — Key Differences Summary Table

Aspect	Fixed Parameters	IES File
Data source	Catalog constants	Measured photometry file
Lux calculation	Lumen method average	Point-by-point inverse-square
Beam angle	Nominal (120° default)	Half-power measured from candela
Uniformity (U_0)	Not available	Calculated from grid
Fixture comparison	Same formula for all	Each product truly different
Asymmetric beams	Not captured	Fully represented
Compliance basis	`lumen method (no IES grid)`	`IES work plane E_avg`
Accuracy	Approximate	Accurate (within IES measurement tolerance)
Speed	Fast (no file I/O)	Slower (grid calculation per layout)

Appendix — Result Fields Reference

Field	Source	Meaning
`Average Lux`	Lumen method	`N × Φ × MF / area`
`E_avg_grid_lx`	IES grid	Mean of all sample point illuminances
`E_min_grid_lx`	IES grid	Lowest sample point illuminance
`E_max_grid_lx`	IES grid	Highest sample point illuminance
`U0_calculated`	IES grid	`E_min / E_avg_grid`
`U1_calculated`	IES grid	`E_min / E_max`
`beam_angle_deg`	IES half-power	Full cone angle at 50% peak intensity
`beam_angle_nominal_deg`	Catalog constant	Always 120° (legacy default)
`Beam source`	Logic flag	`"IES half-power estimate"` or `"nominal catalog value"`
`Standard margin (lux %)`	Compliance	`(E_required - E_avg_grid) / E_required × 100`
`Standard margin (U0 %)`	Compliance	`(U0_required - U0_calculated) / U0_required × 100`
`IES lumens (lm)`	IES header	Manufacturer-stated lumens (or rated if header invalid)
`Maintenance factor`	Settings	Applied MF value (default 0.63)
`Lux compliance basis`	Logic flag	Which average was used for pass/fail